

# **ECEN-635: Homework 3**

Due on September 19, 2025 at 4:30 pm

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## Problem 1

The field  $\mathcal{E}(t) = 5 \sin(20 \times 10^9 t + kz) \hat{\mathbf{y}}$  (V/m) describes a wave propagating in a medium which has the dielectric constant  $\epsilon_r = 3$ . Answer the following questions concerning this wave. Be sure to include dimensions in your answers:

- (a) What is the frequency  $f$  of the wave?
- (b) What is the wave amplitude?
- (c) What is the propagation direction?
- (d) What is the phase velocity of the wave  $v_p$ ?
- (e) What is the wavelength  $\lambda$  of the wave?
- (f) What is the intrinsic impedance of the wave  $\eta$ ?
- (g) What is the wave number (propagation constant)  $k$ ?

## Solution

**Part (a):** The frequency of the wave  $f = \frac{\omega}{2\pi} = \frac{20 \times 10^9}{2\pi} = 3.18 \text{ GHz}$ .

**Part (b):** The wave amplitude of  $\mathcal{E}(t)$  is 5 V/m.

**Part (c):** The propagation direction is in the  $\hat{\mathbf{z}}$  direction.

**Part (d):** The phase velocity  $v_p = \frac{1}{\sqrt{\mu\epsilon}} = \frac{1}{\sqrt{3 \cdot \epsilon_0 \mu_0}} = 1.73 \times 10^8 \text{ m/s}$

**Part (e):** The wavelength of the wave is  $\lambda = \frac{2\pi}{\omega\sqrt{\mu\epsilon}} = \frac{2\pi}{(20 \times 10^9 \text{ rad/s}) \cdot (5.78 \times 10^{-9} \text{ s/m})} = 0.0543 \text{ m}$

**Part (f):** The intrinsic impedance  $\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{3 \cdot \epsilon_0}} = \frac{376.73 \Omega}{\sqrt{3}} = 217.51 \Omega$

**Part (g):** The wave number  $k = \frac{2\pi}{\lambda} = \frac{2\pi}{0.0543 \text{ m}} = 115.71 \text{ m}^{-1}$

## Problem 2

In lossy insulating material a 2.45 GHz plane wave is propagating in a casserole in the  $z$ -direction and polarized in the  $y$ -direction, with an amplitude of 20 V/m, a conductivity  $\sigma = 2.0$  S/m, and a dielectric constant  $\epsilon_r = 4$ . Find the loss tangent. Hint: this is a trivial question but the loss tangent is of primary importance when ordering a material.

## Solution

The loss tangent is given by the formula

$$\tan \delta = \frac{\sigma}{\omega \epsilon}$$

Plugging in the provided values

$$\begin{aligned} \tan \delta &= \frac{2.0 \text{ S/m}}{(2\pi \times 2.45 \times 10^9 \text{ Hz})(4 \cdot \epsilon_0)} \\ &= \frac{2.0 \text{ S/m}}{(2\pi \times 2.45 \times 10^9 \text{ Hz})(4 \cdot 8.854 \text{ pF/m})} \\ &= 3.67 \text{ (dimensionless)} \end{aligned}$$

### Problem 3

The first step in analytically solving for the fields in any situation in electromagnetics is to obtain the vector Helmholtz equation from Maxwell's equations. In class, starting with the Maxwell's equations the electric field vector Helmholtz equation was derived. Starting with Maxwell's equations derive the vector Helmholtz equation for the magnetic field. Write down and number Maxwell's equations and at each step in the derivation tell which equation you are going to use for the next step. If you use a vector identity, before using it write down the identity from those in the back cover of the book. Be sure to use the proper vector symbol and a line above the letter to indicate each vector. You had this problem on the last homework but go ahead and do it again.

### Solution

We have Maxwell's equations:

$$\begin{cases} \nabla \times \mathbf{E} = -j\omega\mu\mathbf{H} & (1) \\ \nabla \times \mathbf{H} = j\omega\epsilon\mathbf{E} & (2) \\ \nabla \cdot \mathbf{E} = 0 & (3) \\ \nabla \cdot \mathbf{H} = 0 & (4) \end{cases}$$

We take the curl of equation (2) to get

$$\nabla \times (\nabla \times \mathbf{H}) = \nabla \times (j\omega\epsilon\mathbf{E}) \quad (5)$$

Then apply the vector identity Appendix II-51 on the left hand side of (5) to get

$$\nabla \times (\nabla \times \mathbf{H}) = \nabla(\nabla \cdot \mathbf{H}) - \nabla^2 \mathbf{H} \quad (6)$$

Next, by substituting (6) into (5), we get

$$\nabla(\nabla \cdot \mathbf{H}) - \nabla^2 \mathbf{H} = \nabla \times (j\omega\epsilon\mathbf{E}) = j\omega\epsilon(\nabla \times \mathbf{E}) \quad (7)$$

Now by applying equations (1) and (4) onto (7), we get

$$-\nabla^2 \mathbf{H} = j\omega\epsilon(-j\omega\mu\mathbf{H}) \quad (8)$$

Simplifying (8) by grouping the constants into  $k^2$  gives us

$$\nabla^2 \mathbf{H} + \omega^2 \mu \epsilon \mathbf{H} = 0 \quad \Rightarrow \quad \nabla^2 \mathbf{H} + k^2 \mathbf{H} = 0 \quad (9)$$

Which is the magnetic field vector Helmholtz equation.

## Problem 4

In each case below (i) show mathematically that the field given is a straight line, a circle or an ellipse (ii) find an expression for the angle  $\psi$  between the field vector and the axis. (iii) In each case describe the type of polarization. Hint: Remember the polarization is found by finding the phase of one of field components with respect to the other. A trick to find the difference in phase is to add or subtract a phase to both field components. By doing this you can cause the first component to have zero phase and the other component will have the difference in the two phases.

- (a)  $\mathbf{E} = (j\hat{\mathbf{x}} + \hat{\mathbf{y}})e^{-jkz}$
- (b)  $\mathbf{E} = [(1+j)\hat{\mathbf{y}} + (1-j)\hat{\mathbf{z}}]e^{-jkx}$
- (c)  $\mathbf{E} = [(2+j)\hat{\mathbf{x}} + (3-j)\hat{\mathbf{z}}]e^{-jky}$
- (d)  $\mathbf{E} = (j\hat{\mathbf{x}} + j2\hat{\mathbf{y}})e^{-jkz}$

Hint: The  $\Psi = 0$  axis is determined by the right hand rule. If the wave travels in the  $z$  direction then  $x \rightarrow y \rightarrow z$  says  $\Psi = 0$  is the  $x$  axis. But if the wave travels in the  $y$  direction then  $z \rightarrow x \rightarrow y$  says  $\Psi = 0$  is the  $z$  axis.

## Solution

**Part (a):** Here  $|E_x| = |E_y| = 1$ , so they have equal amplitudes. Additionally, their phase difference  $\Delta\theta = \pm\pi/2$ , meaning the wave is circularly polarized.

**Part (b):** Now  $|E_y| = |E_z| = \sqrt{2}$ , so they have equal amplitudes. Additionally, their phase difference  $\Delta\theta = \pm\pi/2$ , meaning the wave is circularly polarized.

**Part (c):** Here  $|E_x| = \sqrt{5}$  and  $|E_z| = \sqrt{10}$ . Their phase difference  $\Delta\theta = \arctan(1/2) + \arctan(1/3) = \pi/4$ , meaning the wave is elliptically polarized.

**Part (d):** Here  $|E_x| = 1$  and  $|E_y| = 2$ . Their phase difference  $\Delta\theta = \pi/2 - \pi/2 = 0$ . Thus, the wave is linearly polarized. Then,  $\phi = \arctan|E_y|/|E_x| = \arctan(2) = 63.43^\circ$ .